



Modelling Financial Markets

Seminar 20: Path Signatures and Realised Volatility — Hands-on



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Acronyms used in this material (1/2)

- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- AUDUSD: Australian dollar / US dollar exchange rate (FX pair code)
- BH: Benjamini–Hochberg (procedure)
- BIC: Bayesian Information Criterion
- CI: Confidence Interval
- CL: Crude oil (WTI light sweet) futures, NYMEX ticker
- COVID-19: COronaVirus Disease 2019
- CVX: Chevron Corporation (NYSE ticker)
- DM: Diebold–Mariano (test)
- ES: E-mini S&P 500 futures (CME ticker)
- EURUSD: Euro / US dollar exchange rate (FX pair code)
- FDR: False Discovery Rate
- FWER: Family-Wise Error Rate
- FX: Foreign eXchange
- GBPUSD: British pound / US dollar exchange rate (FX pair code)
- GW: Giacomini–White (test)
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR: Heterogeneous AutoRegressive (model)
- HARQ: HAR model with Realised Quarticity
- JPM: JPMorgan Chase & Co. (NYSE ticker)
- LASSO: Least Absolute Shrinkage and Selection Operator
- MA: Moving Average
- OLS: Ordinary Least Squares
- OPEC: Organization of the Petroleum Exporting Countries
- QLIKE: Quasi-LIKElihood (loss function)

Acronyms used in this material (2/2)

- **RV: Realised Variance**
- **SHAR: Semivariance HAR model**
- **SHW: Sherwin-Williams Company (NYSE ticker)**
- **USDCAD: US dollar / Canadian dollar exchange rate (FX pair code)**
- **USDJPY: US dollar / Japanese yen exchange rate (FX pair code)**
- **VOLARE: VOLatility Archive for Realized Estimates (database)**

Outline

Introduction

Part A: Derivations

Part B: Signatures and Volatility on VOLARE

Part C: Open Problems

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- Compute signatures of piecewise-linear paths step by step and use Chen's identity
- Read the Lévy area as a lead-lag statistic and test whether it predicts volatility
- Compare two volatility forecasts with DM, GW and a bootstrap interval
- Count wins over many assets with Holm and BH corrections

- Background: Lecture 20 and Chapter 9; data: VOLARE [Cipollini et al. \(2026\)](#), free for research at volare.unime.it

Structure

- **Part A:** derivations (A1–A10): signatures, Chen, Lévy area, HAR as AR(22), QLIKE, kernel weights
- **Part B:** VOLARE data with inference (B1–B8)
- **Part C:** a level-aware kernel as project seed (C1); critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with an interpretation question
- Notation: RV_t five-minute realised variance, r_t open-to-close log return,
 $QLIKE(RV, F) = RV/F - \ln(RV/F) - 1$
- Code: [seminar notebook \(Colab\)](#), built on the Quantlet MFM_ch20_seminar; set the folder with your VOLARE files in the first cell

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (signature step by step), A3 (Chen and path reversal), A5 (HAR as AR(22)), A7 (QLIKE), A9 (kernel weights)
- Part B: B1 (Lévy area as predictor, JPM), B3 (HAR, SHAR, HARQ, JPM), B5 (Sig-LK vs log-HAR, JPM), B7 (Holm and BH)

Proposed: you solve them

- Part A: A2 (model A1), A4 (model A3), A6 (model A5), A8 (model A7), A10 (model A9)
- Part B: B2 (model B1, ES), B4 (model B3, CL), B6 (model B5, ES), B8 (model B7)
- Part C: C1, a level-aware kernel; C2, critique an AI answer (model: A1, A8, B5)

A1. A Depth-2 Signature, Step by Step [Solved]

Task

- ▣ A series takes the values $y = (0, 2, 3, 3)$ on days $t = 0, 1, 2, 3$
 - ▣ Assumption: the time-augmented path $X_t = (t, y_t)$ is linear between consecutive days (channel 1 = time, channel 2 = y)
- (a) Write the three increments $\Delta_1, \Delta_2, \Delta_3 \in \mathbb{R}^2$ and compute level 1, $S^{(1)} = \sum_k \Delta_k$
- (b) Use Chen's identity to show that level 2 equals $S^{(2)} = \sum_k \frac{1}{2} \Delta_k \Delta_k^\top + \sum_{j < k} \Delta_j \Delta_k^\top$, then compute the 2×2 matrix
- (c) Check the result with the shuffle identity $S^i S^j = S^{ij} + S^{ji}$ for $i = j = 1$ and for $i = 1, j = 2$
- (d) Compute the Lévy area $A^{12} = (S^{12} - S^{21})/2$
- ▣ **Report:** $S^{(1)}$, $S^{(2)}$ and A^{12} . **Interpretation:** what does the sign of A^{12} say about when the series rose?

A1. Solution (1/2)

Solution

- (a) Increments: $\Delta_1 = (1, 2)$, $\Delta_2 = (1, 1)$, $\Delta_3 = (1, 0)$; level 1: $S^{(1)} = (3, 3)$, the total change in time and in y
- (b) One segment has $\text{Sig} = (1, \Delta, \frac{1}{2}\Delta\Delta^\top, \dots)$; Chen multiplies the segments, and level 2 of the product collects every square term once and every ordered pair $j < k$ once
- Squares: $\frac{1}{2} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1.5 & 1.5 \\ 1.5 & 2.5 \end{bmatrix}$
- Ordered pairs: $\Delta_1\Delta_2^\top + \Delta_1\Delta_3^\top + \Delta_2\Delta_3^\top = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$
- Sum: $S^{(2)} = \begin{bmatrix} 4.5 & 2.5 \\ 6.5 & 4.5 \end{bmatrix}$

A1. Solution (2/2)

Solution

- (c) Shuffle, $i = j = 1$: $S^1 S^1 = 9 = 2S^{11} = 2 \cdot 4.5$; $i = 1, j = 2$: $S^1 S^2 = 9 = S^{12} + S^{21} = 2.5 + 6.5$: both hold
- (d) Lévy area: $A^{12} = (2.5 - 6.5)/2 = -2$
- Geometric check: the path lies above its chord $y = t$ (at $t = 1$: 2 vs 1; at $t = 2$: 3 vs 2); the enclosed polygon has area 2, traversed clockwise, hence the minus sign
- Interpretation: with channels (time, y), a negative area means that y rose *early* in the window; the same net change spread evenly would give $A^{12} = 0$

A2. Signature and Shuffle Identity [Proposed]

Solution discussed in class

Task

- Model: A1. A path in $\mathbb{R}^2$ through the points $(0, 0) \rightarrow (2, 1) \rightarrow (3, -1) \rightarrow (4, 2)$, linear between points
- (a) Write the increments and compute level 1
- (b) Compute level 2 with the formula of A1 (squares plus ordered pairs)
- (c) Verify $S^1 S^2 = S^{12} + S^{21}$ and compute the Lévy area A^{12}
- **Report:** $S^{(1)}$, $S^{(2)}$, A^{12} . **Interpretation:** which channel moved first, on balance?

A3. Chen's Identity and Path Reversal [Solved]

Task

- Path $X : (0, 0) \rightarrow (1, 0) \rightarrow (1, 2)$ from the lecture, with $S^{(1)} = (1, 2)$ and $S^{(2)} = \begin{bmatrix} 1/2 & 2 \\ 0 & 2 \end{bmatrix}$
- $\overleftarrow{X}$ is the same path run backwards: $(1, 2) \rightarrow (1, 0) \rightarrow (0, 0)$
- (a) Compute levels 1 and 2 of $\text{Sig}(\overleftarrow{X})$ with the A1 formula
- (b) Apply Chen's identity, $(A \otimes B)^{(2)} = A^{(2)} + A^{(1)}B^{(1)\top} + B^{(2)}$, to compute level 2 of $X * \overleftarrow{X}$ (there and back)
- (c) State what the result implies for the uniqueness of the signature and for time augmentation
- **Report:** the two signatures and level 2 of the concatenation. **Interpretation:** why do we add a time channel in practice?

A3. Solution

Solution

- (a) Reversed increments: $\overleftarrow{\Delta}_1 = (0, -2)$, $\overleftarrow{\Delta}_2 = (-1, 0)$; level 1: $\overleftarrow{S}^{(1)} = (-1, -2) = -S^{(1)}$
- Level 2: squares $\begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 1/2 & 0 \\ 0 & 0 \end{bmatrix}$ plus the pair $\overleftarrow{\Delta}_1 \overleftarrow{\Delta}_2^\top = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix}$, so $\overleftarrow{S}^{(2)} = \begin{bmatrix} 1/2 & 0 \\ 2 & 2 \end{bmatrix}$
- (b) Chen: $\begin{bmatrix} 1/2 & 2 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ -2 & -4 \end{bmatrix} + \begin{bmatrix} 1/2 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$; level 1 is $S^{(1)} - S^{(1)} = 0$ as well
- (c) There and back is “tree-like”: its signature is that of the constant path; the signature identifies a path only up to such pieces [Hambly & Lyons \(2010\)](#)
- Interpretation: a strictly increasing time channel cannot go back, so tree-like pieces are impossible and the signature becomes unique: this is why we time-augment

A4. Lévy Area of a Loop [Proposed]

Solution discussed in class

Task

- Model: A3. The unit square $(0, 0) \rightarrow (1, 0) \rightarrow (1, 1) \rightarrow (0, 1) \rightarrow (0, 0)$, four linear segments
- (a) Compute level 1 and the Lévy area A^{12} for the square run anticlockwise
- (b) Repeat for the square run clockwise, and for the square translated by $(5, 5)$
- (c) Explain why level 1 cannot distinguish the loop from a constant path while level 2 can
- **Report:** three values of $S^{(1)}$ and A^{12} . **Interpretation:** which invariance does the translation illustrate?

A5. HAR as a Restricted AR(22) [Solved]

Task

- Data: VOLARE Cipollini et al. (2026), JPM (JPMorgan Chase), daily five-minute realised variance rv_5 , 2 Jan 2015 – 30 Jan 2026, trading days only
- Full-sample log-HAR for $\ln RV_{t+1}$ (2 764 usable days), OLS with Newey–West (5 lags) standard errors: $\hat{\beta}_d = 0.411$ (0.026), $\hat{\beta}_w = 0.248$ (0.038), $\hat{\beta}_m = 0.222$ (0.032)
- (a) Substitute the definitions of the weekly and monthly averages into the HAR equation and collect the coefficient of each lag $\ln RV_{t-j+1}$, $j = 1, \dots, 22$
- (b) Compute ϕ_1 , $\phi_2 = \dots = \phi_5$ and $\phi_6 = \dots = \phi_{22}$ from the JPM estimates
- (c) Compute the sum $\sum_{j=1}^{22} \phi_j$
- **Report:** the three distinct values and the sum. **Interpretation:** what does the sum measure, and is the process stationary?

A5. Solution

Solution

- (a) $\beta_w RV_t^{(w)} = \sum_{j=1}^5 (\beta_w/5) RV_{t-j+1}$ and $\beta_m RV_t^{(m)} = \sum_{j=1}^{22} (\beta_m/22) RV_{t-j+1}$ (in logs for log-HAR)
- Lag 1 collects all three terms, lags 2–5 the weekly and monthly ones, lags 6–22 only the monthly one: 22 coefficients, 3 free parameters
- (b) $\phi_1 = 0.411 + 0.248/5 + 0.222/22 = 0.470$
- $\phi_{2..5} = 0.248/5 + 0.222/22 = 0.0596$; $\phi_{6..22} = 0.222/22 = 0.0101$
- (c) $\sum_j \phi_j = \phi_1 + 4\phi_2 + 17\phi_6 = \beta_d + \beta_w + \beta_m = 0.880$
- Interpretation: the sum is the persistence of log RV; $0.880 < 1$, so shocks die out and the process is stationary, but slowly (long-memory-like decay)

A6. Testing the HAR Restrictions [Proposed]

Solution discussed in class

Task

- Model: A5. The unrestricted AR(22) for $\ln RV_t$ has an intercept and 22 slope coefficients $\phi_1, \dots, \phi_{22}$
- Assumption: the regression errors are heteroskedastic and serially correlated
- (a) Count the linear restrictions that HAR imposes on ϕ and write them as $R\phi = 0$
- (b) Write the Wald statistic with an HAC covariance and its distribution under H_0
- (c) State the 5% critical value
- **Interpretation:** if the test rejects in a long sample, should we drop HAR for forecasting?

A7. QLIKE: Minimiser and Asymmetry [Solved]

Task

- $\text{QLIKE}(\text{RV}, F) = \text{RV}/F - \ln(\text{RV}/F) - 1$ with realised proxy $\text{RV} > 0$ and forecast $F > 0$; $\mathbb{E}_t =$ expectation given the information at the forecast origin
- (a) Differentiate $\mathbb{E}_t[\text{QLIKE}(\text{RV}, F)]$ with respect to F and solve the first-order condition
- (b) Check the second-order condition
- (c) For $\text{RV} = 1$, compute QLIKE and the squared error for $F = 0.5$ and for $F = 1.5$
- **Report:** the minimiser and the four losses. **Interpretation:** which error does QLIKE punish more, and why does this suit risk management?

A7. Solution

Solution

- (a) $\mathbb{E}_t[\text{QLIKE}] = \mathbb{E}_t[\text{RV}]/F + \ln F - \mathbb{E}_t[\ln \text{RV}] - 1$; $\partial/\partial F = -\mathbb{E}_t[\text{RV}]/F^2 + 1/F = 0 \Rightarrow F^* = \mathbb{E}_t[\text{RV}]$
- (b) $\partial^2/\partial F^2 = 2\mathbb{E}_t[\text{RV}]/F^3 - 1/F^2$, which equals $1/F^{*2} > 0$ at F^* : a minimum; the same holds for any conditionally unbiased proxy, so the ranking is robust [Patton \(2011\)](#)
- (c) $\text{QLIKE}(1, 0.5) = 2 - \ln 2 - 1 = 0.307$; $\text{QLIKE}(1, 1.5) = 0.667 + \ln 1.5 - 1 = 0.072$; squared errors 0.25 and 0.25
- Interpretation: QLIKE punishes under-prediction of variance about four times more than the same over-prediction; under-predicted risk is the costly error (too little capital, breached VaR)

A8. QLIKE With Its Arguments Swapped [Proposed]

Solution discussed in class

Task

- Model: A7. A program computes $L(RV, F) = F/RV - \ln(F/RV) - 1$, i.e. QLIKE with the arguments swapped
- (a) Find the forecast $F^\dagger$ that minimises $\mathbb{E}_t[L(RV, F)]$
- (b) Compare $F^\dagger$ with $\mathbb{E}_t[RV]$ using Jensen's inequality
- (c) Say which models the swapped loss favours
- **Interpretation:** why is this a silent bug that tests do not catch?

A9. Kernel Weights and Effective Sample Size [Solved]

Task

- ▣ Four past windows with signature distances $\delta = (0.2, 0.5, 1.0, 2.0)$ to the current window
- ▣ Weights of Gu et al. (Eq. 12): $w_\tau = e^{-\gamma\delta_\tau} / \sum_s e^{-\gamma\delta_s}$; Kish effective sample size $n_{\text{eff}} = 1 / \sum_\tau w_\tau^2$
- (a) Compute the four weights for $\gamma = 1$
- (b) Compute the four weights for $\gamma = 3$
- (c) Compute n_{eff} in both cases
- ▣ **Interpretation:** what is gained and what is lost when γ increases?

A9. Solution

Solution

- ▣ (a) $e^{-\delta} = (0.819, 0.607, 0.368, 0.135)$, sum 1.929: $w = (0.425, 0.315, 0.191, 0.070)$
- ▣ (b) $e^{-3\delta} = (0.549, 0.223, 0.050, 0.002)$, sum 0.824: $w = (0.666, 0.271, 0.060, 0.003)$
- ▣ (c) $n_{\text{eff}} = 3.12$ for $\gamma = 1$ and 1.92 for $\gamma = 3$ (out of 4)
- ▣ Interpretation: a higher temperature focuses on the most similar window (less bias from dissimilar periods) but halves the information (more variance); in the lecture $n_{\text{eff}} \approx 841$ of 1000: little is lost, and little changes

A10. Limits of the Temperature [Proposed]

Solution discussed in class

Task

- Model: A9. n past windows, distances $\delta_\tau > 0$, temperature $\gamma = c/\text{median}(\delta)$ as in the lecture
- (a) Find the limits of w and n_{eff} when $c \rightarrow 0$
- (b) Find the limits when $c \rightarrow \infty$ (assume a unique nearest window)
- (c) Show that multiplying all distances by $k > 0$ leaves the weights unchanged
- **Interpretation:** why is the median scaling needed to compare c across 50 assets?

B1. Does the Lévy Area Predict Volatility? (JPM) [Solved]

Task

- Data: VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), daily five-minute realised variance `rv5`, 2 Jan 2015 – 30 Jan 2026, trading days only; open and close prices for the open-to-close return r_t
- Target: $y_t = \ln \text{RV}_{t+1:t+22} - \ln \text{RV}_t^{(m)}$, the change of next-month log RV relative to the past month (overlapping monthly targets)
- (a) For every day, build the path (cumulative r , $\ln \text{RV}$) of the last 22 days and compute its Lévy area A_t from the depth-2 signature; standardise A_t
- (b) Regress y_t by OLS on a constant, the three log-HAR terms and A_t ; compute Newey–West standard errors with 22 lags
- (c) Test H_0 : coefficient of $A_t = 0$ at 5%, two-sided, and compare the R^2 with and without A_t
- **Report:** coefficient, HAC s.e., t , decision, both R^2 . **Interpretation:** why 22 lags, and is the gain economically relevant?

B1. Solution (1/2)

Solution

- (a) Procedure: signature of each 22-point window with channels (return, $\ln RV$), $A_t = (S^{12} - S^{21})/2$; negative in 46% of the windows (leverage-type: prices fell, then volatility rose)
- (b) 2743 usable days; coefficient of $A_t = +0.042$, HAC s.e. 0.036
- (c) $t = +0.042/0.036 = +1.18$; $|t| < 1.96$: do not reject H_0 at 5%
- R^2 : 28.0% without A_t , 28.6% with A_t

B1. Solution (2/2)

Solution

- Why 22 lags: consecutive targets share 21 of their 22 days, so the errors are MA(21) even under H_0 ; fewer lags would understate the standard error
- Relevance: half a point of R^2 next to the 28% explained by the log-HAR terms is negligible
- Conclusion: for JPM the order of moves between price and volatility adds no reliable information about next month's volatility

B2. The Lévy Area for E-mini S&P 500 Futures [Proposed]

Solution discussed in class

Task

- Model: B1. Data: VOLARE, ES (E-mini S&P 500 futures), rv5 and open/close prices, 28 Sep 2009 – 30 Jan 2026
- (a) Repeat steps (a)–(c) of B1 for ES, with the same target, regressors and 22 Newey–West lags
- (b) Test the coefficient of A_t at 1% and 5%, two-sided
- (c) Compare the R^2 gain with the one for JPM
- **Interpretation:** what does a positive coefficient say about windows with a leverage-type (negative) area?

B3. HAR, SHAR and HARQ for JPM, With HAC Errors [Solved]

Task

- Data: VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), daily five-minute realised variance `rv5`, 2 Jan 2015 – 30 Jan 2026, trading days only; semivariances `rsp5`, `rsn5`; quarticity `rq5`
- Target: next-day RV in levels; full sample; Newey–West standard errors with 5 lags
- (a) Estimate HAR [Corsi \(2009\)](#) by OLS and report the three slopes with HAC standard errors
- (b) Estimate SHAR [Patton & Sheppard \(2015\)](#) and test $H_0 : \beta^- = \beta^+$ against $H_1 : \beta^- > \beta^+$ with a Wald t that uses the HAC covariance of the two estimates
- (c) Estimate HARQ [Bollerslev, Patton & Quaedvlieg \(2016\)](#) with $RQ_t^{1/2}$ demeaned and test $H_0 : \beta_Q = 0$ against $\beta_Q < 0$
- **Report:** estimates, t , p , decisions. **Interpretation:** do the two effects of the literature hold for this stock?

B3. Solution (1/2)

Solution

- (a) HAR (2 764 days): $\beta_d = 0.367$ (0.080), $\beta_w = 0.494$ (0.149), $\beta_m = -0.026$ (0.095)
- (b) SHAR: $\beta^+ = 0.401$ (0.252), $\beta^- = 0.324$ (0.236); $\hat{\beta}^- - \hat{\beta}^+ = -0.078$
- $t = (\hat{\beta}^- - \hat{\beta}^+) / \sqrt{V_{--} + V_{++} - 2V_{+-}} = -0.17$, one-sided $p = 0.566$: do not reject H_0
- (c) HARQ: $\hat{\beta}_Q = -35.062$ (15.904), $t = -2.20 < -1.645$: reject H_0 at 5% one-sided

B3. Solution (2/2)

Solution

- Reading HARQ: on a day with high quarticity (noisy RV) the effective daily coefficient $\beta_d + \beta_Q(RQ_t^{1/2} - \overline{RQ^{1/2}})$ falls: the model trusts that day's RV less
- Reading SHAR: the standard errors of β^+ and β^- are large because the two semivariances are highly correlated for a single stock
- Conclusion: the HARQ effect holds for JPM; the good/bad volatility asymmetry is not detectable for JPM alone (it is a panel result in Patton and Sheppard)

B4. HAR, SHAR and HARQ for Crude Oil [Proposed]

Solution discussed in class

Task

- Model: B3. Data: VOLARE, CL (crude oil futures), rv5, rsp5, rsn5, rq5, 28 Sep 2009 – 30 Jan 2026
- (a) Estimate SHAR with Newey–West (5 lags) and test $\beta^- > \beta^+$ as in B3(b)
- (b) Estimate HARQ and test $\beta_Q < 0$ as in B3(c)
- (c) Compare both results with JPM
- **Interpretation:** why might the asymmetry be stronger for a commodity than for a bank stock?

B5. Sig-LK vs log-HAR for JPM, One Day Ahead [Solved]

Task

- Data: VOLARE Cipollini et al. (2026), JPM (JPMorgan Chase), daily five-minute realised variance $rv5$, 2 Jan 2015 – 30 Jan 2026, trading days only
- Design of the lecture: rolling window of 1000 pairs, re-estimation every 5 days, path (time, $\ln RV$, cumulative r) over 22 days, depth 3, two-step LASSO with BIC, $c = 0.5$
- (a) Produce one-day forecasts of Sig-LK and log-HAR from 27 Jan 2020 to 29 Jan 2026 (1 511 days) and compute the mean QLIKE of each
- (b) Compute the ratio of mean QLIKE and a 95% CI with the stationary bootstrap (mean block length 10, 1000 replications)
- (c) Run the DM test (HAC variance, bandwidth $\max(h, \lceil T^{1/3} \rceil)$) and the conditional GW test with instruments $(1, d_{t-1})$
- **Report:** losses, ratio and CI, t , p , decisions. **Interpretation:** is there any evidence that the signature adds value for JPM?

B5. Solution

Solution

- (a) Mean QLIKE: Sig-LK 0.1726, log-HAR 0.1727; on average 4.2 regressors selected, effective sample size 851 of 1000
- (b) Ratio 1.000; bootstrap 95% CI [0.956, 1.047] contains 1
- (c) DM: $d_t = \text{QLIKE}_t^{\text{Sig-LK}} - \text{QLIKE}_t^{\text{log-HAR}}$, $t = \bar{d} / \sqrt{\hat{\Omega}_{\text{NW}} / T} = -0.02$, one-sided $p = 0.493$: do not reject equal accuracy
- GW: Wald statistic $2.85 \sim \chi^2(2)$, $p = 0.241$: the loss difference is not predictable either
- Interpretation: no evidence; the two forecasts are practically identical, and the CI rules out gains or losses larger than about 5%

B5. Cumulative Loss Difference (JPM, One Day Ahead)



- Upward segments: log-HAR better; downward segments: Sig-LK better; the largest moves sit in March 2020 and April 2025
- A handful of stress days decides the sign of the total: report the cumulative path, not only the mean

B6. Sig-LK vs log-HAR for ES, One Week Ahead [Proposed]

Solution discussed in class

Task

- Model: B5. Data: VOLARE, ES (E-mini S&P 500 futures), rv_5 and open/close prices, 28 Sep 2009 – 30 Jan 2026; horizon $h = 5$, $c = 0.5$
- (a) Produce weekly forecasts (average RV over the next 5 days) of Sig-LK and log-HAR with the design of B5
- (b) Compute the QLIKE ratio and its stationary-bootstrap 95% CI (mean block length 10)
- (c) Run the DM test (bandwidth at least $h = 5$) and the conditional GW test
- **Interpretation:** why is the CI so asymmetric, and what does it imply for the power of DM?

B7. Holm and BH, Step by Step [Solved]

Task

- From the lecture: one-sided DM p -values for “Sig-LK better than log-HAR” at $h = 1$ (QLIKE), one per asset; family = the 50 assets
- The five FX pairs, sorted: USDJPY 0.213, EURUSD 0.218, USDCAD 0.246, GBPUSD 0.678, AUDUSD 0.688
- (a) Apply Holm (FWER 5%) to the five FX p -values: compare $p_{(k)}$ with $0.05/(m - k + 1)$ and stop at the first non-rejection
- (b) Apply BH (FDR 5%) to the same five: find the largest k with $p_{(k)} \leq 0.05 k/m$
- (c) Repeat both procedures for all 50 assets and count the rejections
- **Report:** rejections raw, Holm, BH. **Interpretation:** what may we claim, and about how many assets?

B7. Solution

Solution

- (a) Holm, $m = 5$: $p_{(1)} = 0.213$ vs $0.05/5 = 0.01$: no rejection, stop; all Holm-adjusted $p = 1$
- (b) BH: thresholds 0.01, 0.02, 0.03, 0.04, 0.05; every $p_{(k)}$ is above its threshold: no rejection; BH-adjusted $p = 0.410$ for the first three
- (c) 50 assets: 2 raw rejections; Holm: the smallest, ES with $p < 0.001$, is below $0.05/50 = 0.001$: rejected; $p_{(2)} = 0.028 > 0.05/49$: stop, so 1; BH: 1
- Interpretation: we may claim an improvement for one futures contract (ES); nothing for FX or for stocks as a group, and even ES needs a new, pre-registered confirmation

B8. Multiple Testing at One Month Against HAR [Proposed]

Solution discussed in class

Task

- ▣ Model: B7. From the lecture: 50 one-sided DM p -values for “Sig-LK better than HAR” at $h = 22$ (QLIKE)
- (a) Count the rejections at 5% without correction
- (b) Apply Holm and BH as in B7 and count the rejections; name the asset with the smallest p -value
- (c) Compare with the pooled mean QLIKE ratio of Sig-LK relative to HAR at $h = 22$ (1.332)
- ▣ **Interpretation:** how can Sig-LK beat HAR on several assets and still lose on average?

C1. A Level-Aware Signature Kernel [Proposed]

Solution discussed in class

Task

- Context: the kernel of the lecture is translation-invariant, so it cannot recognise a high-volatility regime (Lecture 20, weights on 16 March 2020); data: VOLARE, the 40 American stocks, $rv5$ and open/close prices, 2015–2026
- Idea: prepend a **basepoint** [Morrill et al. \(2020\)](#): the path starts at the training mean of $\ln RV$ and jumps to the first value of the window; by Chen, $\text{Sig} = \exp_{\otimes}(v) \otimes \text{Sig}(X)$ with v the standardised level jump
- (a) Write a one-page pre-registration: 40 stocks, $h = 1$, the lecture design, $c = 0.5$, QLIKE, DM with Holm, the COVID-19 subperiod reported separately
- (b) Implement the basepoint kernel (use the Chen update of the notebook) with the same features as Sig-LK, and forecast
- (c) Compare with Sig-LK and log-HAR: mean QLIKE ratios (all days and COVID-19), wins over log-HAR raw and after Holm
- **Interpretation:** if seeing the level does not help, where would you look next (features, target, horizon)?

C2. Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **Student prompt:** “rv holds the daily 5-minute RV of one stock. Build the signature LASSO of Gu et al. for next-day log RV and evaluate it with QLIKE on the last 30% of the sample.”

- **Answer of the AI assistant**

```

lrv = np.log(rv.values)
S = np.array([sig_path(lrv[t-21:t+1, None], depth=3)      # 22-day window
              for t in range(21, len(lrv) - 1)])
y = lrv[22:]                                              # next-day log RV
lam = LassoCV(cv=5).fit(S, y).alpha_                     # choose lambda
cut = int(0.7 * len(y))
m = Lasso(alpha=lam).fit(S[:cut], y[:cut])
f = np.exp(m.predict(S[cut:]))
qlike = lambda a, b: a / b - np.log(a / b) - 1
print(qlike(f, np.exp(y[cut:])).mean())

```

- “The signature LASSO has a lower QLIKE than HAR on the test set, so path signatures capture the shape of volatility and improve the forecast.”

C2. Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

Task

- ▣ Context: the answer on the previous slide contains **three** errors that are typical of signature and volatility code; the rest (loading, the split, the exponential back-transform) is correct
- ▣ Model: A1 and A3 (time augmentation), A8 (swapped QLIKE), B5 (evaluation without look-ahead)
- (a) For each error, name the line and explain in one sentence why it is wrong
- (b) For each error, describe a check that would reveal it without knowing the answer (a unit test, a timing test, a known case)
- (c) Write the corrected code and compare it with log-HAR on the same split, with a DM test
- (d) Log each error as in `AI_ERRORS.md`: what the AI answered, what was wrong, how you found out, the correct version
- ▣ **Interpretation**: does the conclusion of the assistant follow even if the code were correct?

Common Pitfalls

- ▣ Signatures of one-dimensional paths, or without the time channel
- ▣ Hyperparameters (λ , c , depth) tuned on data that are later used for evaluation
- ▣ QLIKE with swapped arguments; DM tests without HAC variance for $h > 1$
- ▣ Comparing only with HAR in levels, so that the log transform gets credited to the new method
- ▣ Reporting the best asset or horizon instead of the family, corrected with Holm or BH

Take-Home Assignment

- ▣ Solve the proposed problems A2, A4, A6, A8, A10 and B2, B4, B6, B8 in the [seminar notebook](#)
- ▣ C1: write a one-page pre-registration for your own variant of the level-aware kernel and run it on five assets
- ▣ C2: correct the AI code and compare it with log-HAR on the same split
 - ▶ Log every AI error you catch from now on in [AI_ERRORS.md](#), as in C2: what the AI answered, what was wrong, how you found out, the correct version

Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Statistics of Financial Markets
- ☐  Applied Time Series Analysis with Python

Code and data

- ☐ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_20; data: volare.unime.it

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